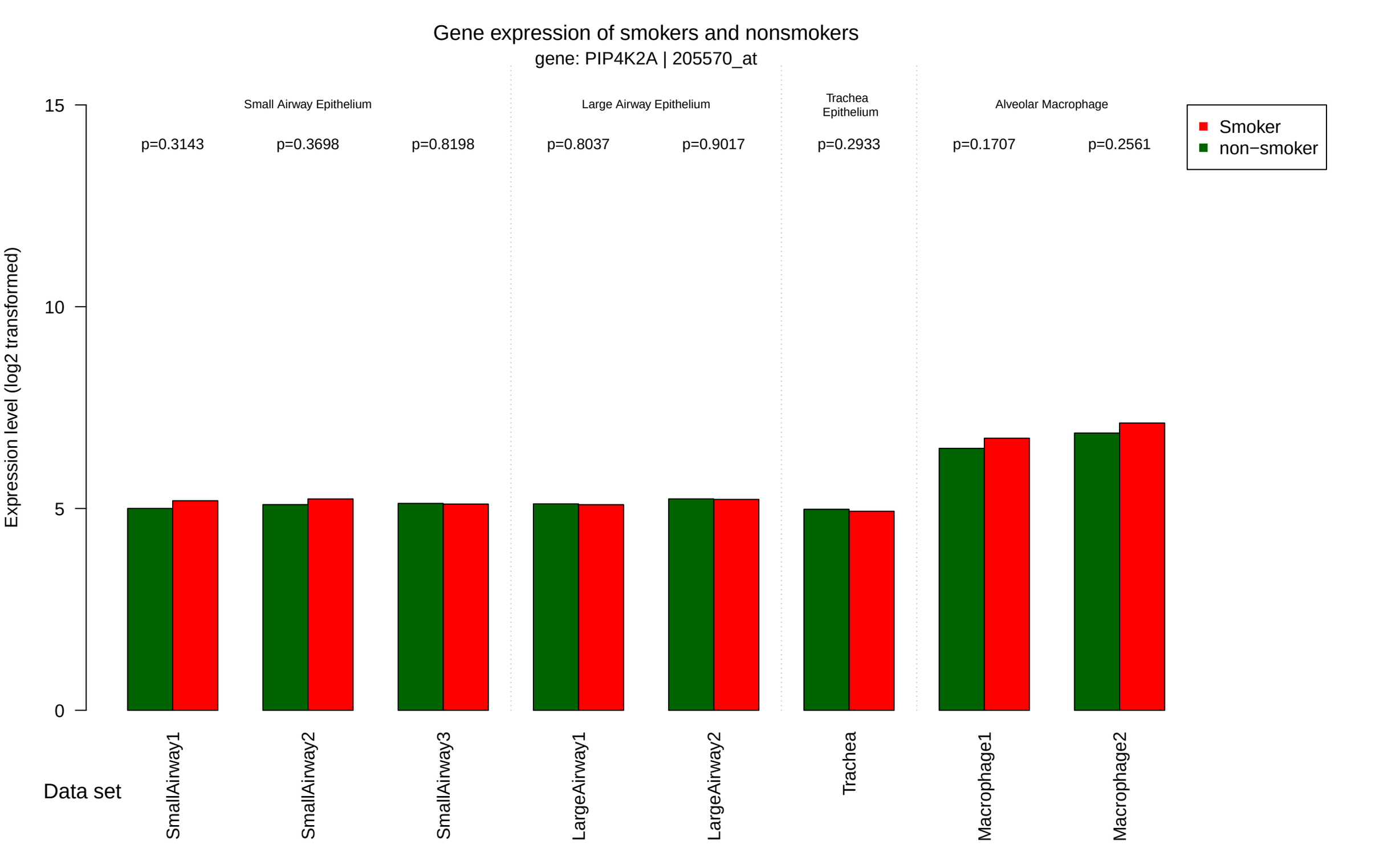
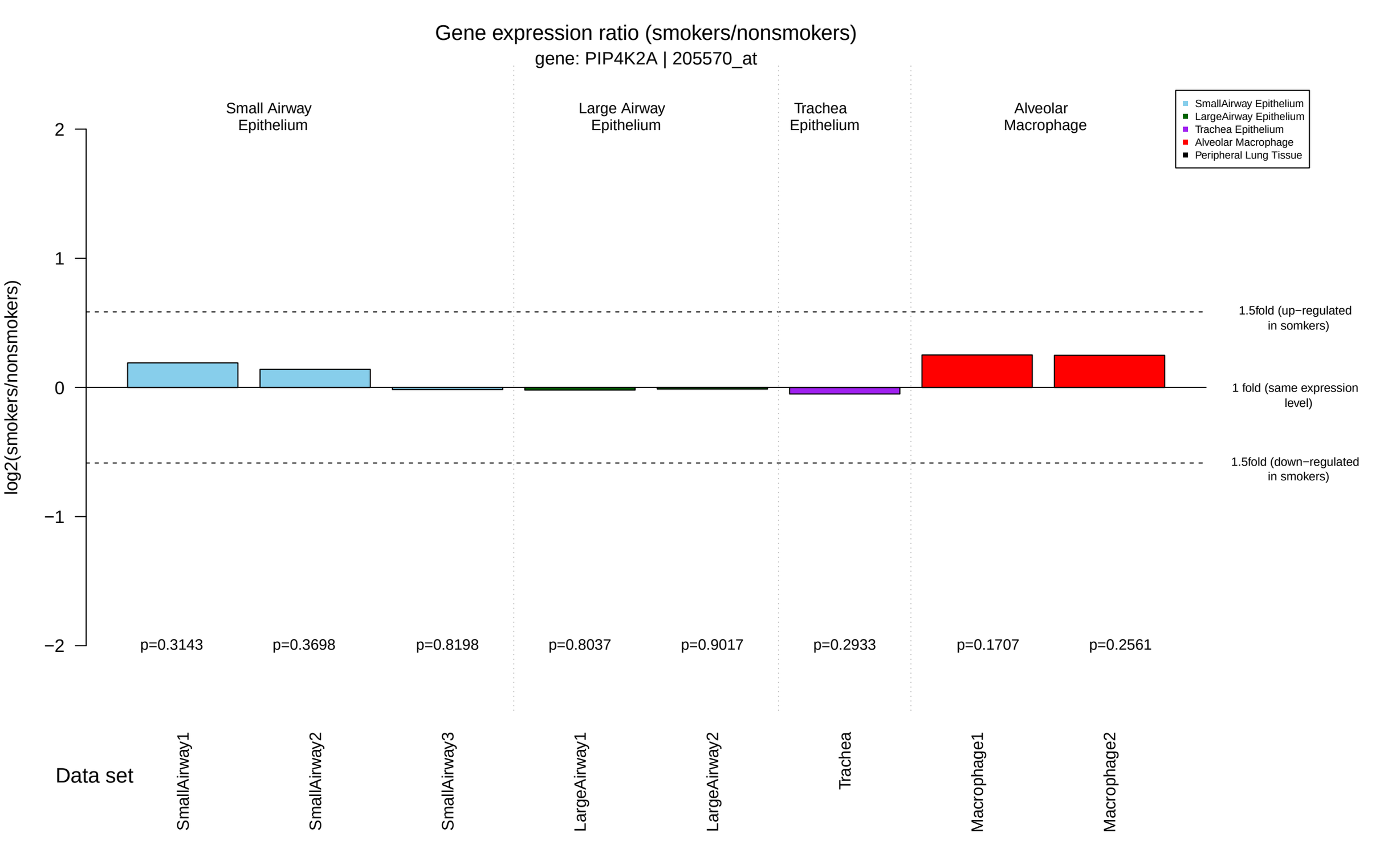
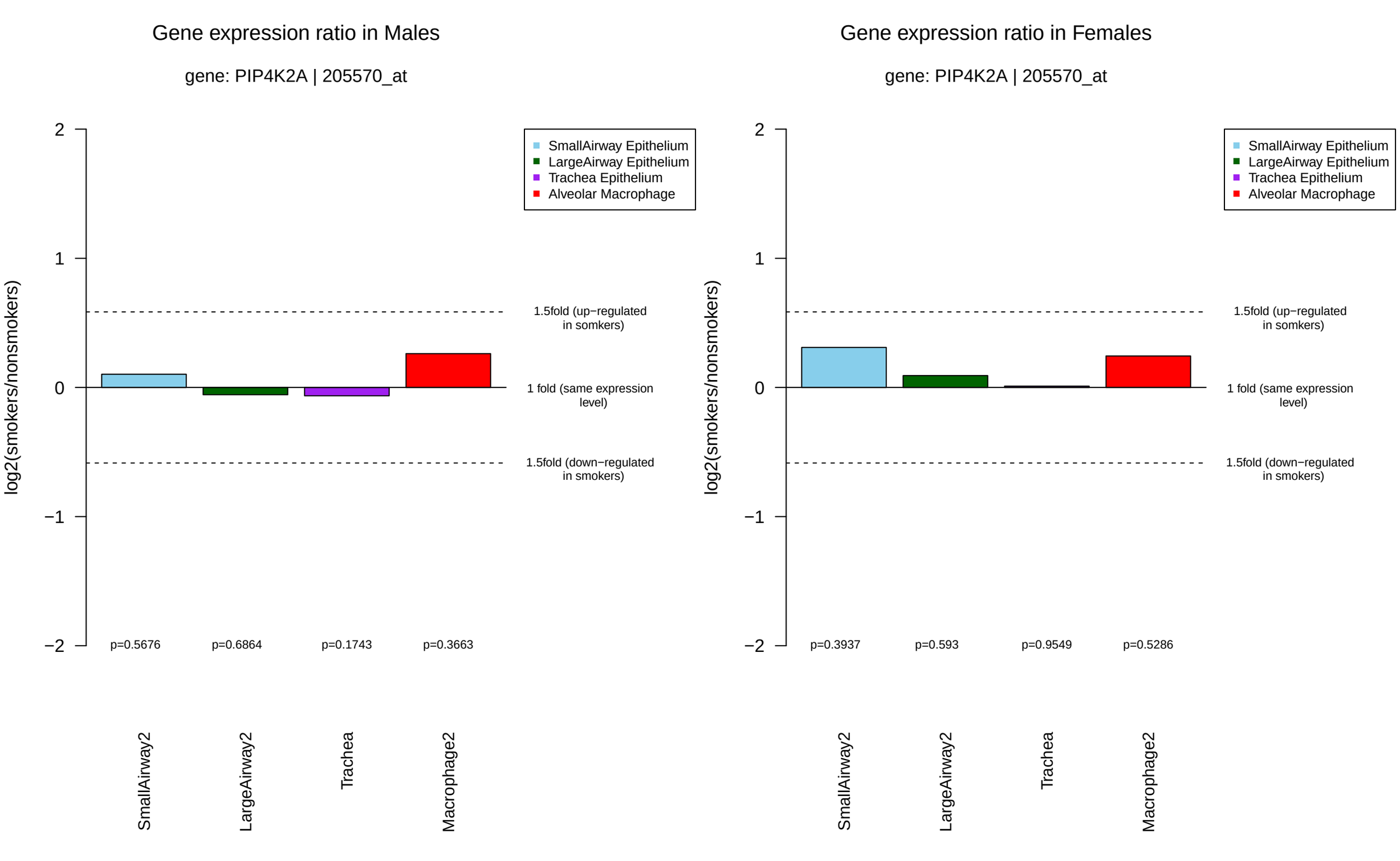
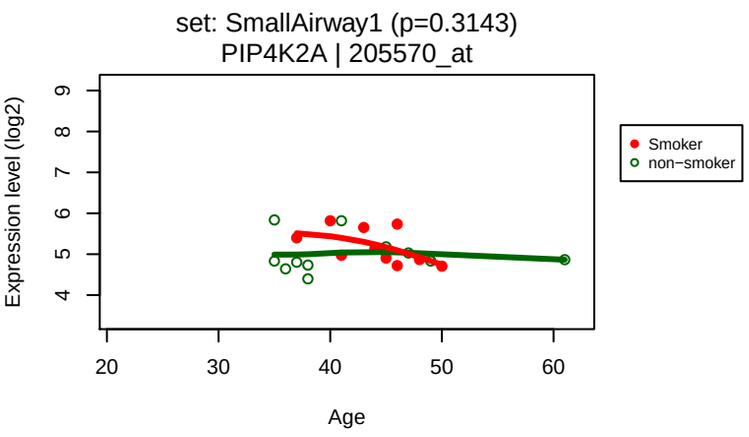
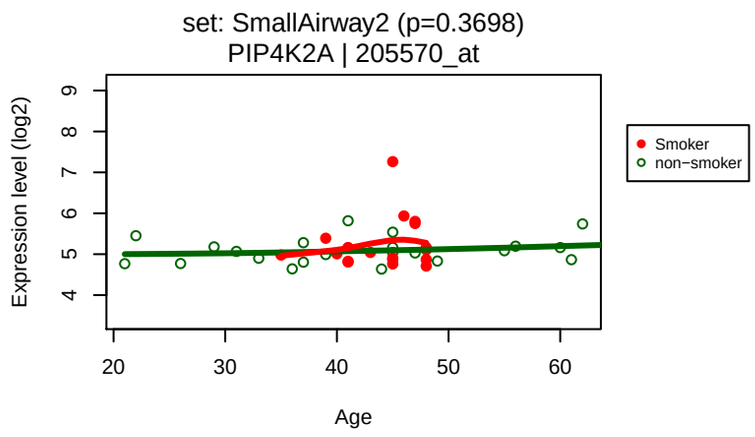
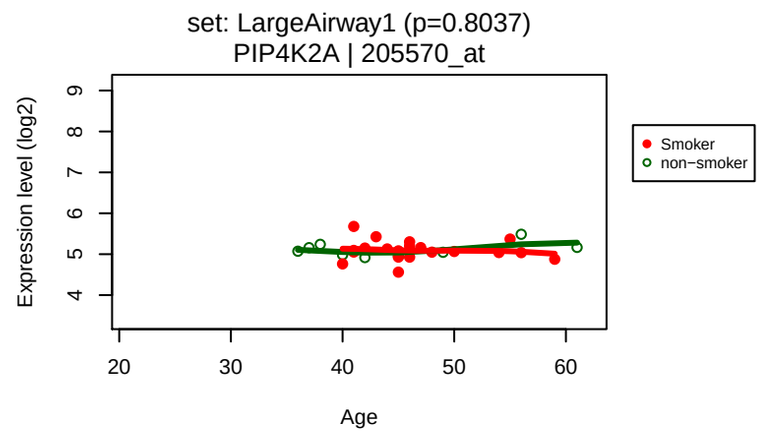

Gene expression of smokers and nonsmokers
gene: PIP4K2A | 205570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3143
p=0.3698
p=0.8198
p=0.8037
p=0.9017
p=0.2933
p=0.1707
p=0.2561
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

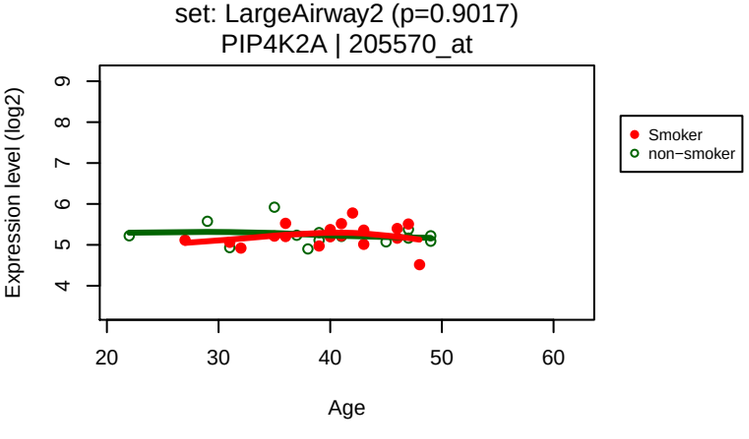

Gene expression ratio (smokers/nonsmokers)
gene: PIP4K2A | 205570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3143
p=0.3698
p=0.8198
p=0.8037
p=0.9017
p=0.2933
p=0.1707
p=0.2561
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

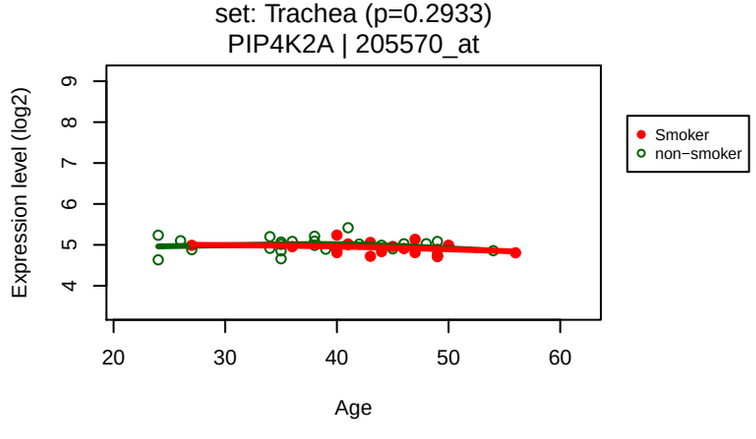

Gene expression ratio in Males
gene: PIP4K2A | 205570_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5676
p=0.6864
p=0.1743
p=0.3663
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIP4K2A | 205570_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3937
p=0.593
p=0.9549
p=0.5286
SmallAirway2
LargeAirway2
Trachea
Macrophage2

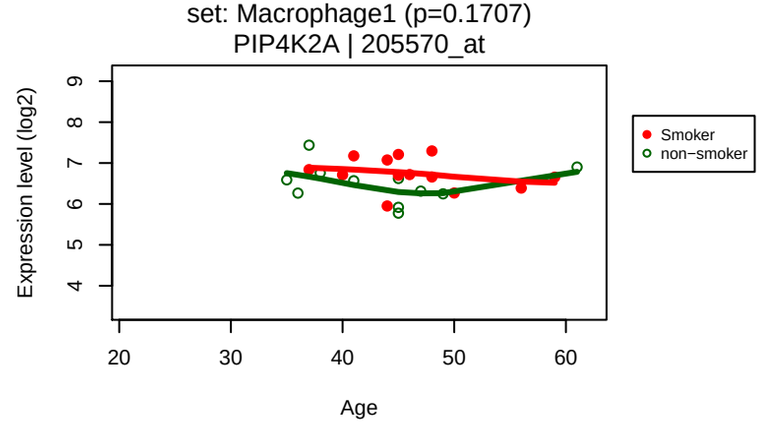

set: SmallAirway1 (p=0.3143)
PIP4K2A | 205570_at
Expression level (log2)
Age
Smoker
non−smoker

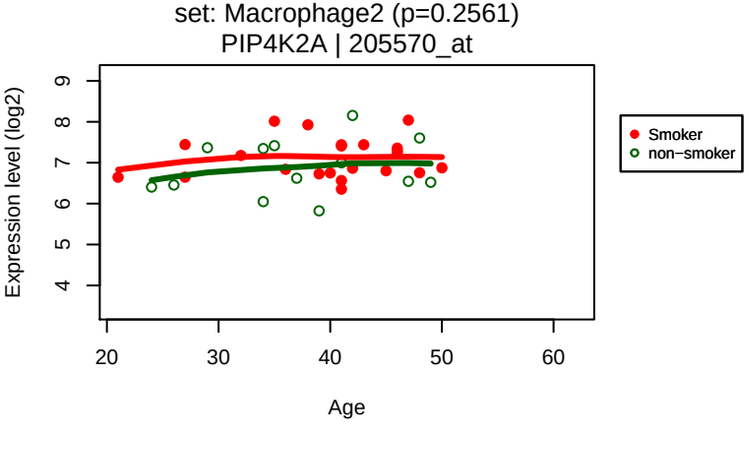

set: SmallAirway2 (p=0.3698)
PIP4K2A | 205570_at
Expression level (log2)
Age
Smoker
non−smoker

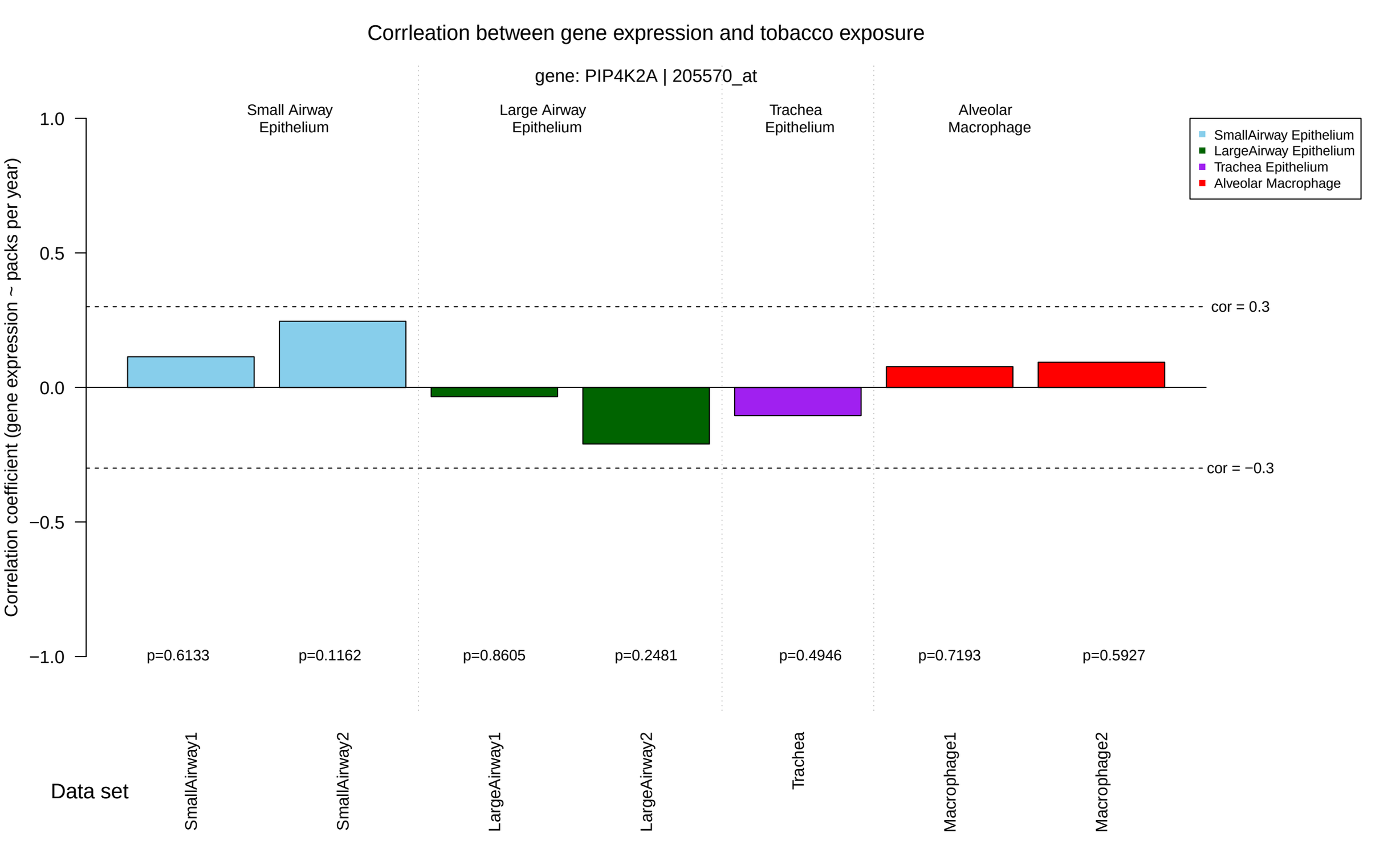

set: LargeAirway1 (p=0.8037)
PIP4K2A | 205570_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9017)
PIP4K2A | 205570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2933)
PIP4K2A | 205570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1707)
PIP4K2A | 205570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2561)
PIP4K2A | 205570_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIP4K2A | 205570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6133
p=0.1162
p=0.8605
p=0.2481
p=0.4946
p=0.7193
p=0.5927
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2